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Accurate determination of polyethylene pellet density using transmission Raman spectroscopy

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In this study, transmission Raman spectroscopy was explored for the direct measurement of the density of packed polyethylene (PE) pellets. A simple and direct transmission Raman measurement of packed, solid granules or pellet samples without pretreatment is greatly advantageous. Initially, the optimal packing thickness of PE pellets for transmission Raman measurement was determined by investigating the reproducibility of triplicate spectra collected by varying the thickness from 2 to 9 cm. Once determined, transmission Raman spectra were collected for 25 different grades of PE pellets and the partial least squares method was used to determine the sample density. The resulting accuracy was 0.00067 g·cm^{-3}, while that obtained using backscattering measurements was 0.00083 g·cm^{-3}. To investigate possible inhomogeneity within a pellet, Raman line mapping was performed over the face of a sectioned pellet and spectral variations among the mapped spectra were examined using principal component analysis. In addition, differential scanning calorimetry was performed on three samples prepared separately by cutting a pellet into left, middle, and right sections. Based on both studies, internal pellet inhomogeneity was found to be minute, but was clearly present. The correct sample representation of internally inhomogeneous PE pellets by the transmission Raman measurement eventually improved the accuracy for density determination. Finally sample-to-sample two-dimensional correlation analysis was used to further examine the origin of the improved accuracy. **Copyright © 2011 John Wiley & Sons, Ltd.**

Keywords: transmission Raman measurement; polyethylene pellets; density; internal inhomogeneity; sample-to-sample two-dimensional correlation

Introduction

When ordinary backscattering Raman spectroscopy is used for the analysis of solid samples that are inhomogeneous in composition, or are composed of multilayers such as coated pharmaceutical tablets, the obtained Raman features are more representative of the sample surface rather than the whole sample identity. In a situation like this, it is necessary to improve the sample representation. Recently, transmission Raman spectroscopy[1–10] and spatially offset Raman scattering (SORS)[7–13] have been proposed as alternatives that can reduce the subsampling problem compared to the conventional backscattering Raman measurement. Transmission Raman is based on the collection of Raman signals from the opposite side of laser illumination so that Raman spectra representing the bulk content of samples can be acquired. SORS involves the collection of Raman spectra from regions spatially offset on the sample surface from the laser beam interaction point, thereby leading to a wider spread of photons interacting with deeper sample layers. The demonstrated applications of transmission Raman or SORS are mostly based on the measurement of a single solid sample with apparent inhomogeneity (e.g. coated pharmaceutical tablets).

To increase transmission Raman or SORS applicability, these methods should be able to directly measure packed pellets or granule samples, which are the most frequently encountered solid sample presentation for nondestructive spectroscopic analysis in diverse fields such as polymers, agriculture, and feed industry. For a nondestructive and fast analysis of these types of samples, diffuse reflectance near-infrared (NIR) spectroscopy[14,15] has been mostly adopted, although its spectral feature could be dominated by information pertinent to the sample surface. Typically, these samples are packed in a large container and the corresponding NIR spectrum is obtained by rotating or reciprocating the samples during spectral collection to improve reproducibility and sample representation. Therefore, if reliable transmission Raman measurements of packed granular samples are feasible, the more informative Raman features could be utilized to improve measurement accuracy. So far, no studies showing the capability of transmission Raman spectroscopy for direct analysis of packed solid granules or pellets have been demonstrated.

In this study, we explore the use of transmission Raman spectroscopy for the direct measurement of density in packed polyethylene (PE) pellets. An optimal packing thickness for transmission measurements was initially investigated, the transmission spectra were collected, the density of PE pellets was determined using partial least squares (PLS),[16,17] and the resulting accuracy was compared with that from backscattering measurement. In

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addition, the spectral features obtained from both transmission and backscattering measurements were closely compared.

Inhomogeneity within a pellet was investigated using Raman microscopic line mapping across the surface of a sliced single pellet, and a series of mapped spectra were carefully examined to observe tendencies indicating inhomogeneity. If inhomogeneity exists within a pellet, the use of backscattering measurements would not be recommended, especially for quantitative analysis. To further confirm possible inhomogeneity within a pellet, differential scanning calorimetry (DSC) was employed. DSC thermograms of three sections, prepared by cutting a pellet into left, middle, and right parts, were examined. Finally, to explain the difference in accuracy between backscattering and transmission measurements, sample-to-sample two-dimensional (2D) correlation analysis[18–20] was used to examine the dissimilarity among the sample spectra in each dataset.

Experimental

Samples and collection of transmission Raman spectra

Twenty-five different types of polyethylene (LLDPE: linear low-density polyethylene) pellets that were used in a previous publication[21] were also used in this study. The samples were of three different grades; 1-butene copolymerized PEs (12 samples), 1-octene copolymerized PEs (9 samples) and homo PEs (4 samples). Generally, the lower density pellets had a more disk-like shape with a larger diameter (approximately 5 to 6 mm), while the higher density pellets were more spherical with a smaller diameter (approximately 4 to 5 mm).

Transmission Raman spectra were collected by directly illuminating the PE pellet packing that was housed in a rectangular glass container by laser light of 785 nm (Innovative Photonic Solutions, Monmouth Junction, NJ, USA). The diameter of illuminated spot was 1.5 mm approximately and the power of laser illumination at the samples was 380 mW. Transmitted Raman scattering was collected at the opposite side of laser illumination using a wide area illumination (WAI) scheme (PhAT system, Kaiser Optical Inc., Ann Arbor, MI, USA), which was the same instrumentation as used to collect the backscattered Raman signals of the same PE pellets in a previous publication.[21] The beam collection area was 28.3 mm^2. Triplicate spectra were collected for each sample. At the time of each spectrum collection, PE pellets were reloaded into the glass container, ensuring random pellet packing for each measurement. The sample thickness was varied, ranging from 2 to 9 cm, to study the optimal packing thickness for transmission measurements. Transmission Raman spectra were collected with an exposure time of 4 s for 50 scans (spectral resolution: 4 cm^{-1}).

All calculations, including baseline correction, intensity normalization, principal component analysis (PCA), PLS regression, and 2D correlation analysis were conducted using Matlab version 7.0 (The Math-Works Inc., MA, USA).

Raman mapping of a pellet

For Raman mapping inside a PE pellet, only a thin oval-shaped slice corresponding to the middle section of a pellet was obtained by cutting out both outer sections. The thickness of PE slice was approximately 1.2 mm. The thin slice was simply positioned on a slide glass and Raman spectra were collected in series across the sectioned surface using a dispersive Raman microscope (Kaiser Optical Inc., $\lambda_{ex} = 785$ nm). The sectioned sample was positioned on a microscope stage and the laser beam (approximately 10 μm in diameter) was focused using an objective lens (10×/0.25 numerical aperture) to collect the Raman spectra across the sectioned face. The point-to-point distance between each Raman spectral collection was 45 μm.

DSC analysis

To examine the internal property gradient inside PE pellets as observed in the Raman mapping, we also investigated the property variations using DSC (differential scanning calorimeter), which is a standard method to analyze the thermal behavior of a polymer. For this purpose, the middle section of a high-density pellet was sliced out to obtain an oval-shaped slice. Then, the top and bottom portion of the slice was removed to get a nearly rectangular shape with the width and thickness of approximately 3.5 and 0.5 mm, respectively. This slice was evenly divided into three sections; left, middle, and right sections. The above procedure was repeated for eight different pellets of the same density to collect sufficient sample quantity for DSC analysis. Finally, DSC analysis was performed using the accumulated sections from each location (DSC Q-100, TA Instrument Corp., New Castle, DE, USA). The thermograms were measured with a heating rate of 10 °C/min from 30 to 150 °C under a flow of nitrogen (50 ml/min).

Results and Discussion

Determination of an optimal packing thickness for transmission Raman measurement

The effort to find an optimal packing thickness for transmission Raman measurement of the PE pellets was based on the assumption that the pellet packing thickness could influence the reproducibility of the collected spectra. Three samples of low, middle, and high density were selected and the corresponding transmission Raman spectra were collected at various packing thicknesses ranging from 2 to 9 cm in 1 cm increments. The spectra were collected by laser excitation for 4 s with an accumulation of 50 scans (total acquisition time: 200 s).

Figure 1 shows the transmission Raman spectra of the low-density sample (density: 0.918 $g{\cdot}cm^{-3}$) collected with different packing thicknesses over the 1504–1054 cm^{-1} range. The baselines were linearly corrected at 1504 and 1054 cm^{-1}. Assignments of the bands in the figure can be found in our previous publication.[21] Raman intensities of the PE band were strongest at 2 cm packing thickness and gradually decreased with increasing packing thickness. To quantitatively present the decrease of Raman intensity, the intensity of the 1298 cm^{-1} band was plotted at each packing thickness (inset, Fig. 1). The absolute decrease in Raman intensity was larger for thin and smaller for samples that were more thickly packed. When the logarithm of the intensity at 1298 cm^{-1} *versus* the packing thickness was plotted, a linear relationship was observed, indicating that the percent intensity decrease per centimeter was constant regardless of the packing thickness. The generated Raman signals, as well as illuminating laser radiation, are attenuated due to scattering, reflection, and refraction as they propagate through a thick pellet packing. Therefore, the use of a smaller packing thickness should be advantageous since spectra of higher signal-to-noise ratio could be acquired at a given spectral collection period. The same plots as shown in Fig. 1 for the middle (0.941 $g{\cdot}cm^{-3}$) and high (0.961 $g{\cdot}cm^{-3}$) density samples were examined, although the corresponding figures are not shown. The

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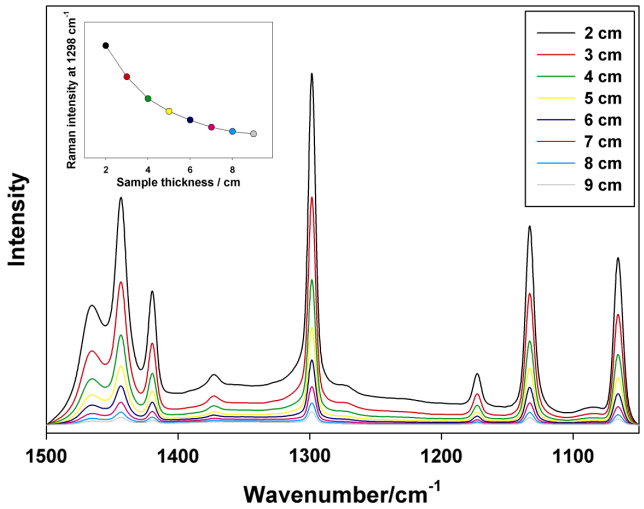

Figure 1. Transmission Raman spectra of low-density PE pellets (density: 0.918 g·cm^{-3}) collected with different packing thicknesses. Baselines of spectra were corrected at 1504 and 1054 cm^{-1}. The band intensity at 1298 cm^{-1} is plotted at each packing thickness (inset).

trend of spectral variations for these samples was almost similar as that observed in Fig. 1.

Packing thickness could also influence the reproducibility of the collected spectra, which is also an important factor governing the accuracy of transmission measurements. Since PE pellets are granular and their size is fairly large, packing is not always consistent, and considerable void space is present inside the packing. When a laser beam hits the pellets, it can then diffuse in different directions. Most of the diffused laser radiation should fully interact with other pellets to generate the Raman signal. However, a tiny portion of radiation could propagate through the void space inside the packing by successive reflections at pellet surfaces and/or partially interact with only the outer surfaces of a pellet. Therefore, the sample representation of collected spectra could vary because of pellet packing as well as packing thickness.

To evaluate the reproducibility of spectral collection at different packing thicknesses, triplicate spectra were collected for the same three PE samples stated above by varying the packing thickness. For each triplicate collection, the pellets were removed from the cell and reloaded to ensure random packing. As discussed earlier, the signal-to-noise ratio of spectra decreases with increase in packing thickness; therefore, it is reasonable to control the signal-to-noise ratio of collected spectra as close as possible for a fair comparison of the reproducibility. For this purpose, the Raman intensity collected at each packing thickness was intentionally controlled to be as similar to each other as possible by changing the number of spectral accumulations with a fixed exposure time of 4 s. For example, to measure the 9 cm packing, the scan number was increased to 360 to generate a similar Raman intensity as at the 2 cm packing, which was acquired with only 13 scans.

Figure 2(a) shows the normalized transmission Raman spectra in the 1504–1054 cm^{-1} range of the low-density PE pellet (density: 0.918 g·cm^{-3}) with different packing thicknesses. For normalization, the baselines were linearly corrected at 1504, 1405, 1349, 1255, 1195, 1112, and 1054 cm^{-1} and the area under the 1504–1054 cm^{-1} range was calculated. Each spectrum was divided by the corresponding area for normalization. Based on visual comparison of the spectra, the spectral features were nearly the same regardless of the packing thickness.

To highlight any possible unseen difference in the triplicate spectra, difference spectra were obtained by subtracting each triplicate spectrum one by one. In this way, three difference spectra were obtained for each measurement. Figure 2(b)–(d) show the resulting difference spectra at each packing thickness for the low-, middle-, and high-density PE samples, respectively. The difference spectra were arbitrarily offset for the comparison. For the low- and middle-density samples, the magnitude of difference was relatively small up to 5 cm packing, with a subsequent increase in difference from 6 to 9 cm packing. When the packing thickness was larger, such as 6–9 cm, laser radiation and the generated Raman signal traveled longer through the pellet packing, eventually leading to more uncertainty in their propagation inside of packing. As shown, the spectral difference was relatively large where the Raman bands were located (Fig. 2).

Relatively reproducible spectra were collected for the high-density sample at each packing thickness. The better reproducibility was presumably related to sample packing homogeneity. To visually observe the packing homogeneity, pellet packing images for the three samples are shown in Fig. 3. The shape of the high-density pellet was nearly spherical, while that of the middle- and low-density samples were more disk-like. It was expected that the spherically shaped pellet would be more homogeneously packed than the disk-like pellet. As shown, the high-density pellets were more or less evenly packed (Fig. 3). In the middle- and low-density samples, horizontally packed regions were observed with a larger void space, thus resulting in less homogeneous pellet packing. Based on the results in Figs 2 and 3, we conclude that homogeneity of the pellet packing influences the reproducibility of transmission Raman measurements.

To quantitatively evaluate the spectral differences among the triplicate spectra, linear regression was performed on each difference spectrum to determine the line of best fit, and standard deviation relative to the straight line was calculated. For each packing thickness, a total of nine calculated standard deviation values from three different samples (low-, middle-, and high-density) were averaged (Fig. 4). As expected, the magnitude of the averaged standard deviation was large for thick PE pellet packing and small for thin PE pellet packing.

It is difficult definitively explain the observation of optimal thickness at 4 cm compared to 2 or 3 cm. Presumably, the occurrence of non-fully interacted radiation inside the void volume would be minimized at 4 cm and the corresponding sample representation would be satisfactory by sufficient interaction of laser radiation with the pellets. Consequently, 4 cm packing was chosen for the transmission Raman measurement of PE pellets. Transmission Raman spectra for all 25 samples were collected at this packing thickness for further quantitative density analysis.

Transmission and backscattering Raman spectra of PE pellets

Figure 5(a) and (b) show the backscattering and transmission Raman spectra (1504–1054 cm^{-1} range) collected from all 25 PE pellet samples. Only one spectrum from triplicate spectra of each sample is displayed. Based on a simple visual comparison of the spectra, the overall features from both measurements were quite similar. In both cases, the Raman spectra from the three samples in the low-density range (shown in green color) show apparently different backgrounds due to weak fluorescence. One sample from the 1-butene grade and two samples from the 1-octene grade showed underlying weak fluorescence backgrounds, similar to that observed in an earlier study.[21]

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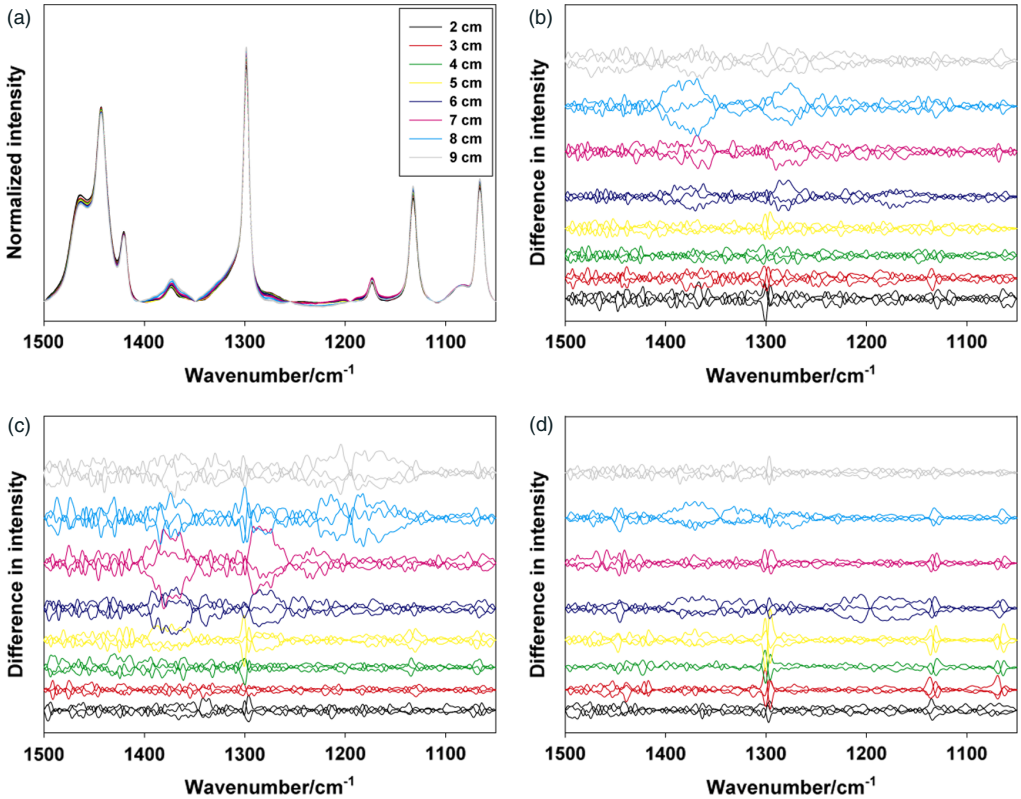

Figure 2. Normalized transmission Raman spectra of low-density PE pellets (density: 0.918 g·cm^{-3}) with different packing thicknesses in the 1504–1054 cm^{-1} range. Difference spectra at each packing thickness for low- (b), middle- (c), and high-density PE (d) samples are shown. The same color notation is used for all plots.

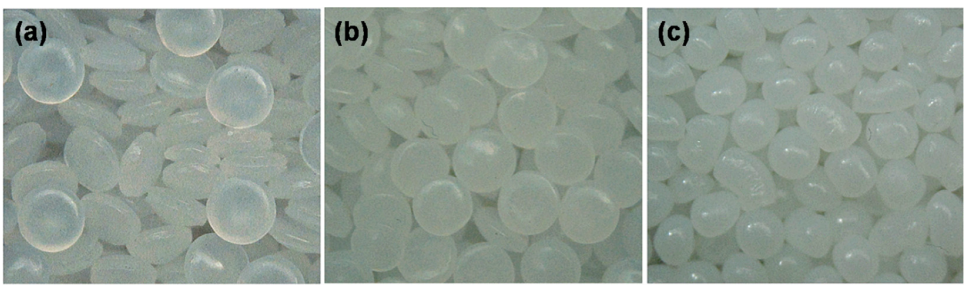

Figure 3. Pellet packing images for low- (a), middle- (b), and high-density (c) PE samples.

The underlying baselines in the transmission and backscattering spectra were slightly different. Except for the three spectra superimposed with the fluorescence background, the baselines of backscattering spectra were nearly flat across the 1504–1054 cm^{-1} range while those of the transmission spectra were slightly tilted. The tilted baseline in the transmission measurement was due to scattering of the Raman signal during propagation through packed pellets. Since the degree of scattering is a function of wavelength (proportional to the fourth power of frequency), relatively large scattering will occur in the lower wavenumber region (closer to the excitation wavelength) and the resulting intensity will decrease as a result of the increased scattering. The generated Raman signals in the higher wavenumber range (further from excitation wavelength), such as the 1504–1400 cm^{-1} range, were less scattered, so the baseline intensities were relatively high compared to those in the lower wavenumber region. For easier understanding, the *x*-axis was also expressed as absolute wavelength of the Raman scattering on top of each figure. When transmission collection was used, the resulting baselines had slowly declining slopes from high to low wavenumbers. In addition, the baselines were more vertically varied. This clearly indicates the variability in the propagation of radiation through the sample, which is due to different pellet packing characteristics at each measurement.

Figure 6(a) shows the normalized backscattering and transmission Raman spectra of low-density PE pellets (density: 0.918 g·cm^{-3}) in the 1504–1054 cm^{-1} range. Only one spectrum from triplicate spectra is shown for each measurement. For closer comparison, the 1475–1415, 1306–1290, and 1150–1054 cm^{-1} ranges are enlarged in Fig. 6(b)–(d), respectively. After normalization, the peak intensities of transmission spectra, especially for the 1298, 1130, and 1070 cm^{-1} bands, were relatively high compared to those of the backscattering spectra. Since the intensities of these bands were somewhat lower than in their raw transmission

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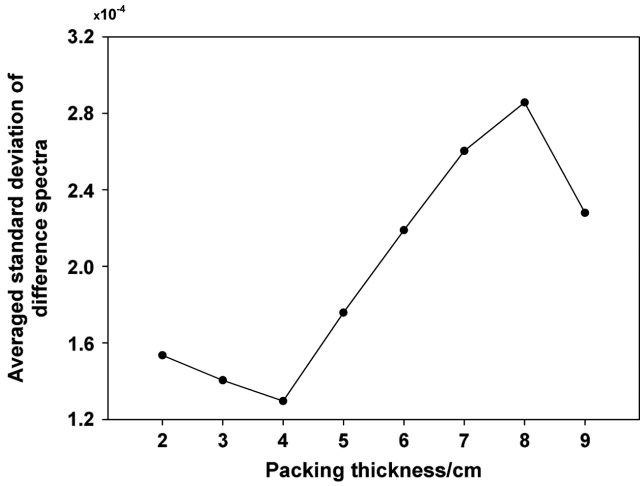

Figure 4. Averaged standard deviations of difference spectra as a function of packing thickness.

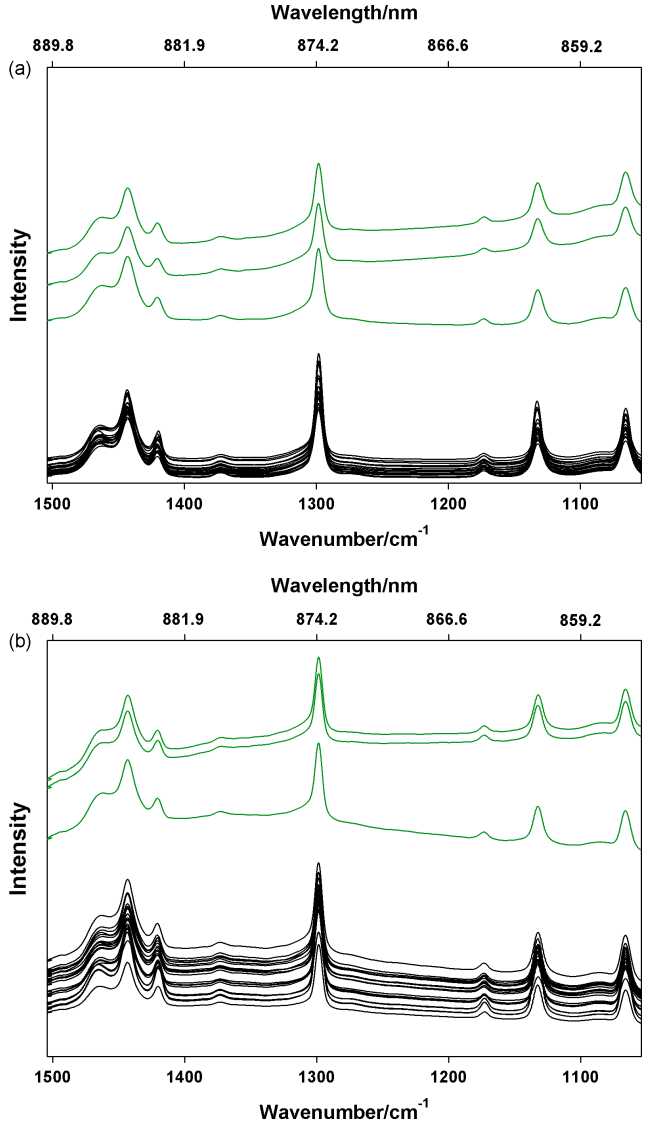

Figure 5. Backscattering (a) and transmission Raman spectra (b) collected from all 25 PE pellet samples. One spectrum from triplicate spectra of each sample is displayed. Three samples in the low-density range showing weak fluorescence are shown in green color.

spectra because of a larger degree of scattering, the intensities of these bands slightly increased during the course of normalization. Meanwhile, the normalized intensities of the three Raman bands at 1462, 1442, and 1421 cm^{-1} in the 1504–1400 cm^{-1} range from both measurements were nearly similar (Fig. 6(b)).

The spectral features between the backscattering and transmission spectra in the 1475–1415 and 1150–1054 cm^{-1} range were almost similar in terms of band shape and position. However, the bandwidth at 1298 cm^{-1} was slightly narrower when the transmission scheme was employed (Fig. 6(c)). The 1298 cm^{-1} band corresponds to the C–C backbone stretching. The narrower band implies a more rigid backbone structure. The spectral difference indicates that sample representation of the low-density PE pellets, either by backscattering or transmission are slightly different.

Backscattering and transmission Raman spectra of high-density PE pellets (density: 0.961 $g{\cdot}cm^{-3}$) were also examined. Figure 7(a) shows both spectra in the full spectral range. Similarly, the 1475–1415, 1306–1290, and 1150–1504 cm^{-1} ranges are enlarged in Fig. 7(b)–(d), respectively. In the high-density PE pellet, the spectral features between the backscattering and transmission measurement were more different, especially in terms of the bandwidth. Except for the 1465 cm^{-1} band, the bandwidths of the other five bands at 1442, 1419, 1298, 1133, and 1065 cm^{-1} in the transmission spectrum were slightly narrower compared to those in the backscattering spectrum.

Ideally, the spectral features of PE pellets acquired from both measurements should be nearly identical if the PE pellet is homogeneous. If the molecular structure or composition between the outside and inside of a pellet is different, backscattering measurements could potentially produce spectral features that would be more representative of the outer part of the pellet. Investigation of possible inhomogeneity within a single pellet was necessary to address the spectral differences between backscattering and transmission spectra of PE pellets, especially for high-density pellets as shown in Fig. 7.

PLS regression to determine PE pellet density

Partial least squares[16,17] regression was used to determine the density of PE pellets using the transmission measurements, and the resulting accuracy was compared with that obtained from the backscattering measurements. Normalized spectra were directly used for PLS without further pretreatments such as mean-centering and variance-scaling. As performed in a previous study,[21] the resulting error was obtained by cross-validation with a set of five validation samples. Initially, 20 and 5 samples were assigned to the calibration and validation sets, respectively. There are 53 130 combinations when selecting five validation samples from a dataset of 25 samples. All the combinations were tested, and the mean standard error of calibration (MSEC) and the mean standard error of prediction (MSEP) were calculated by averaging the individual SEC and SEP values. The whole spectral range (1504–1054 cm^{-1}) was used for PLS. The resulting MSEP resulting from the transmission measurement was 0.00067 $g{\cdot}cm^{-3}$ with the use of six factors, while the MSEP obtained from backscattering measurement was 0.00083 $g{\cdot}cm^{-3}$ with the use of four factors.[21] The improvement in accuracy was statistically significant based on a *t*-test at the 95% confidence level ($P_{calculated} = 0.02 < 0.05 = P_{95\%}$). Since the dataset was fully cross-validated to obtain the result, the improved accuracy achieved by the transmission

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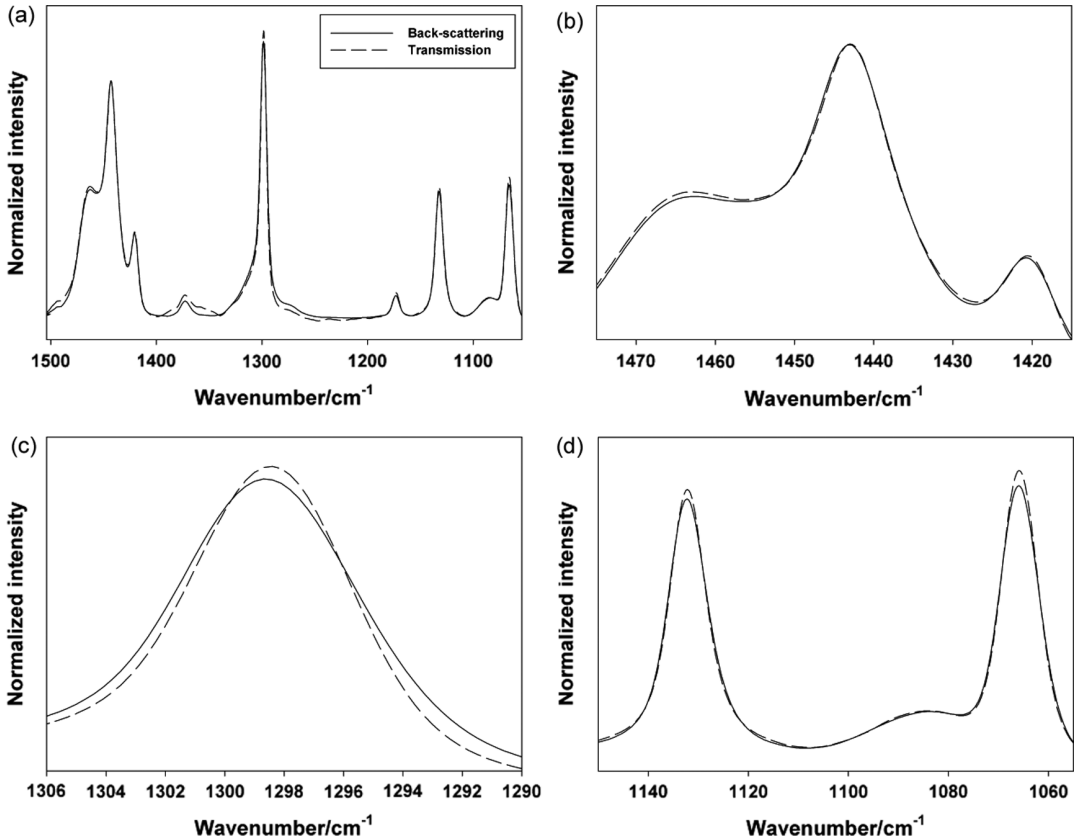

Figure 6. Normalized backscattering and transmission Raman spectra of low-density PE pellets (density: 0.918 g cm^{-3}) in the 1504–1054 cm^{-1} range (a). For closer comparison, the 1475–1415 (b), 1306–1290 (c), and 1150–1054 cm^{-1} (d) range are enlarged. The same line notation is used for all plots.

measurement is statistically significant. Meanwhile, to assess the practical applicability of the proposed transmission measurement in an actual field, it would be better to further validate the transmission measurement with the use of an independent dataset in long run.

Raman mapping of single pellet

To examine possible internal inhomogeneity of a pellet, Raman line mapping was performed across a sectioned pellet. By moving the sectioned face from left to right, Raman spectra were regularly collected with a point-to-point spacing of 45 µm. The laser spot was approximately 6 to 8 µm in diameter. The low- and high-density pellets used in Figs 6 and 7, respectively, were used for line mapping. Nearly identical Raman spectral features acquired from line mapping are indicative of internal pellet homogeneity. Conversely, internal inhomogeneity would be confirmed if spectral variation, either systematic or random, is observed among the line mapping spectra. A total of 97 and 67 spectra were collected for the low- and high-density pellets, respectively. A larger number of spectra were collected for the low-density pellet since its size was larger. To easily visualize the variation among the collected spectra, PCA[16,17] was performed using each dataset and the variation of first scores along the mapping line was investigated. As performed in PLS, normalized spectra in the 1504–1054 cm^{-1} range were used for PCA without further spectral pretreatments.

Figure 8(a) shows the variation of the first scores across the mapping line (mapping length: approximately 5 mm) for the low-density PE pellet. The numbers in the figure correspond to the serial numbering of the mapping points from left to right. Although the score variations were not dramatic, it was possible to observe an approximate U-shaped variation. This is an interesting result supportive of the molecular structural differences between the outside and inside of a pellet. To examine the actual spectral variation, four spectra (#1 from the left edge, #30 and #54 from the middle, and #97 from the right edge) were selected and designated in Fig. 8(a), which are shown Fig. 8(b). The spectral features of the four selected spectra were quite similar over the whole spectral range; however, a rather systematic variation was observed when the 1442 and 1171 cm^{-1} bands were magnified (Fig. 8(c) and (d), respectively). The band shape and position of the spectra collected at both edges (#1 and #97) were relatively close to each other, while those of spectra collected in the middle (#30 and #54) were also relatively close. Although the spectral differences between the outside and inside of a pellet were not significant, it was sufficient to demonstrate a trend for micro-inhomogeneity inside a pellet.

Figure 9(a) shows the variation of the first scores across the mapping line (mapping length: approximately 3.5 mm) for the high-density PE pellet. In the case of the high-density pellet, the variation of scores was a more distinct U-shape. Five selected spectra (#1 and #67 from both edges, #35 from the center, #9 and #61 from the middle) in the 1504–1054 cm^{-1} range are shown in Fig. 9(b). For detailed examination, the 1442 and 1298 cm^{-1} bands are highlighted in Fig. 9(c) and (d), respectively. The features of the selected spectra depended on the spectral collection points on the pellet. Variation of the 1442 cm^{-1} band among these five spectra was distinct; its intensity was highest and lowest at both edges

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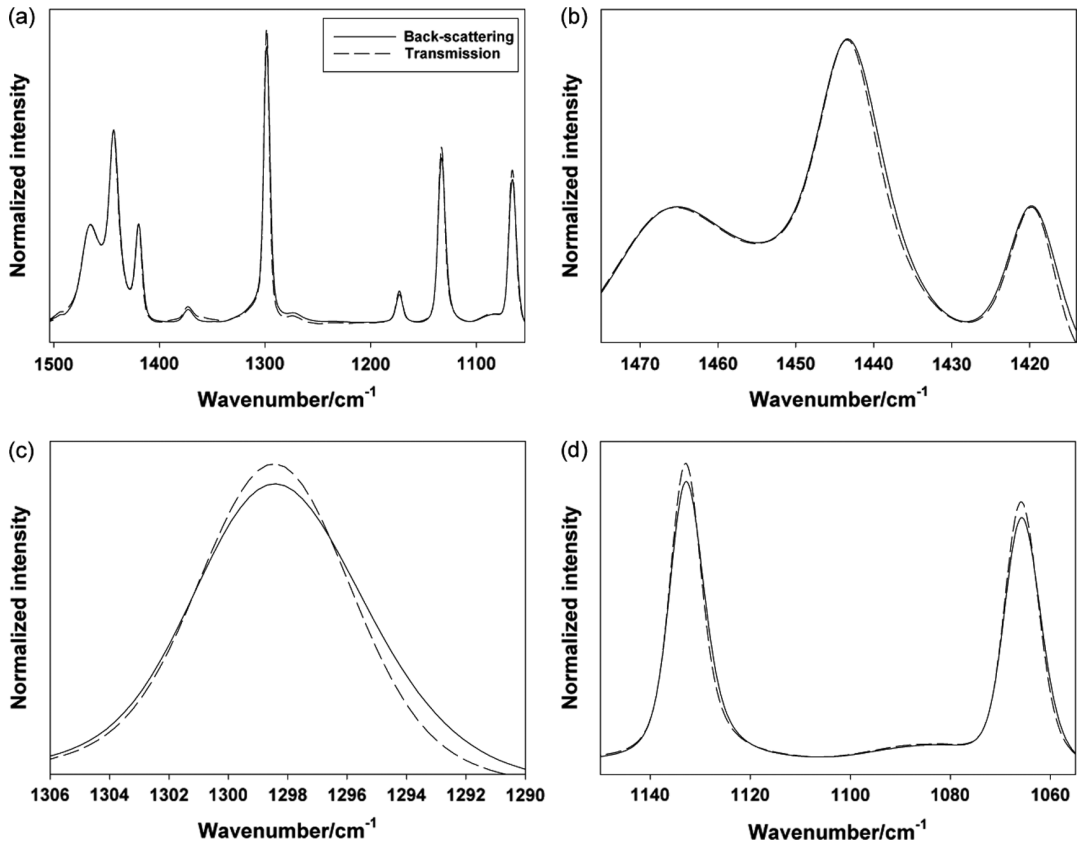

Figure 7. Normalized backscattering and transmission Raman spectra of high-density PE pellets (density: 0.961 g cm^{-3}) in the 1504–1054 cm^{-1} range (a). For closer comparison, the 1475–1415 (b), 1306–1290 (c), and 1150–1504 cm^{-1} (d) range are enlarged. The same line notation is used for all plots.

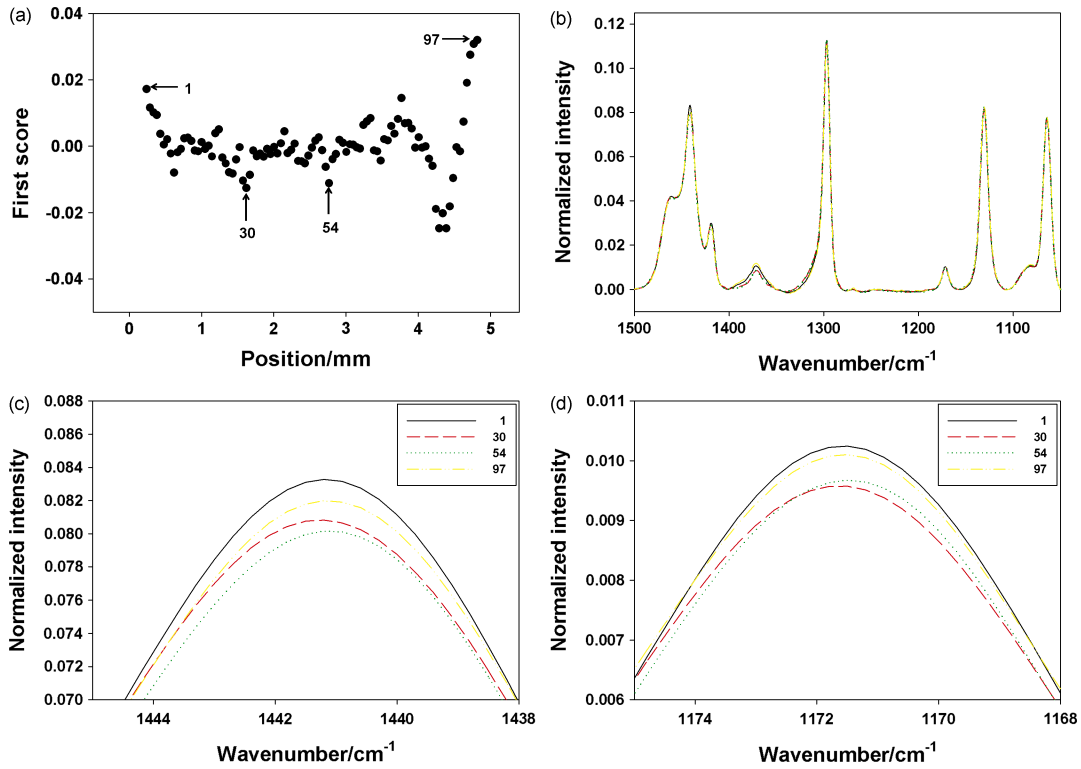

Figure 8. Variation of first scores across the mapping line for low-density PE pellets (a) (density: 0.918 g·cm^{-3}). The selected spectra are shown in the 1504–1054 (b), 1445–1438 (c), and 1175–1168 cm^{-1} (d) range. The spectra of black, red, green and yellow color were collected at the mapping position of #1, 30, 54 and 97, respectively.

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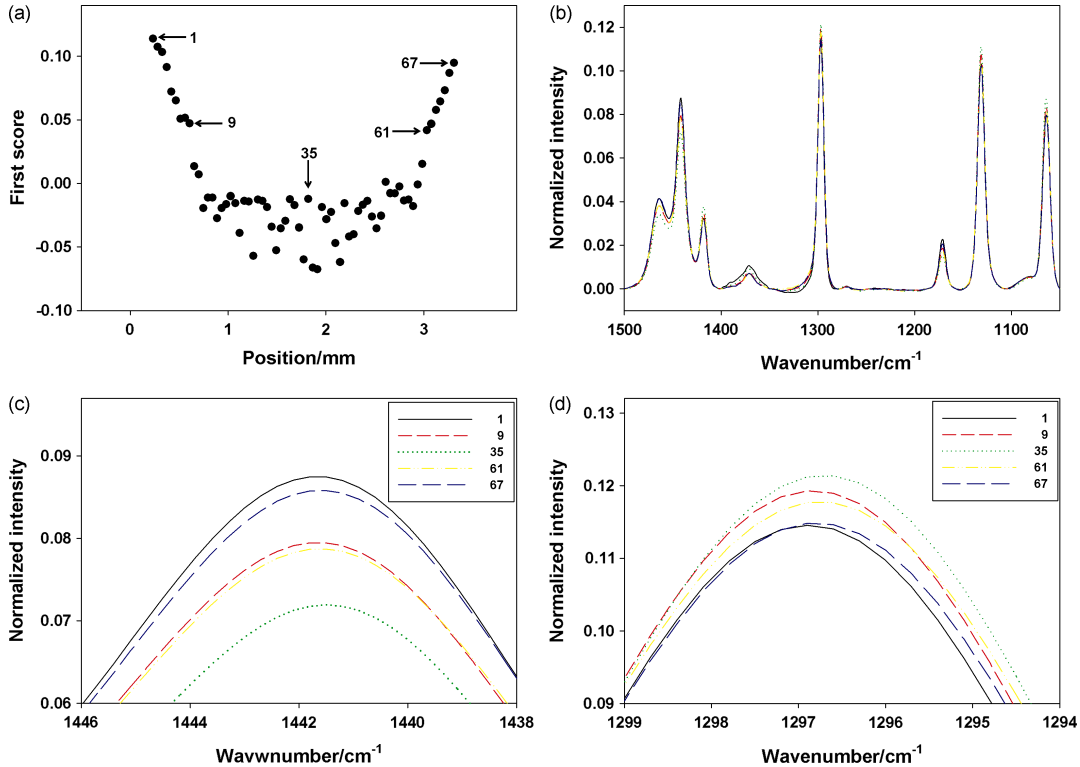

Figure 9. Variation of first scores across the mapping line for high-density PE pellets (a) (density: 0.961 g cm^{-3}). The selected spectra are shown in the 1504–1054 (b), 1446–1437 (c), and 1299–1294 cm^{-1} (d) range. The spectra of black, red, green, yellow and blue color were collected at the mapping position of #1, 9, 35, 61 and 67, respectively.

and the center, respectively (Fig. 9(c)), while the spectral features of the two spectra collected at both edges were quite similar. In addition, the two spectra obtained at both middle points were close to each other.

As shown in Fig. 9(d), the band shape and position of the spectra collected at the edge, middle, and center were slightly different, while the two spectra collected at both edges were similar to each other and the two spectra obtained at both middle points were also analogous. Overall, the Raman mapping results clearly demonstrate that there is gradual property variation even within a single pellet, indicating pellet inhomogeneity. The high-density pellets had more distinct internal property variation.

Identification of internal inhomogeneity using DSC analysis

Figure 10 shows the thermograms acquired from the three sections of high-density pellet samples. For detailed examination, the 125–135 °C range corresponding to melting temperature (T_m) is highlighted in the figure. The melting temperatures of the left and right sections were almost similar around 132 °C, while that of the center section was approximately 0.5 °C higher. The higher T_m indicates a more rigid polymeric structure. Although DSC results were obtained from only three sections because of the small pellet size, the data are in accordance with the results obtained from Raman line mapping, which clearly indicates inhomogeneity as well as property gradients within a single pellet. The inhomogeneity could occur at the pelletizing process where a molten polymer is cut into individual pellets and allowed to cool under ambient conditions. During the cooling process, the outside of the pellet starts to cool earlier than the inside. This difference in

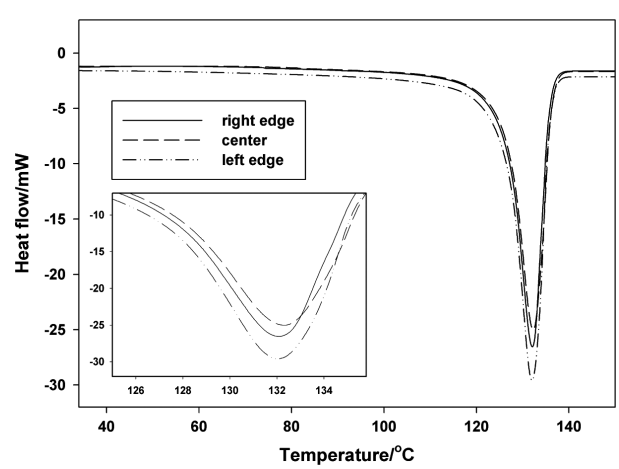

Figure 10. DSC thermograms acquired from three sections of high-density pellet samples (density: 0.961 g cm^{-3}). The 125–135 °C range corresponding to melting temperature (T_m) is highlighted inside the figure.

cooling rate, although minute, could introduce property gradients in the final pellet.

Although Raman information pertinent to the inside of a pellet can be also obtained using backscattering collection, the resulting spectra should be more representative of the outer portion of a pellet based on a previous publication, in which the intensity of Raman scattering as a function of penetration depth was demonstrated in a solid sample using Monte-Carlo simulation.[1] Since density at the center of a pellet is slightly higher, as determined by DSC analysis (Fig. 10), the narrower PE bands

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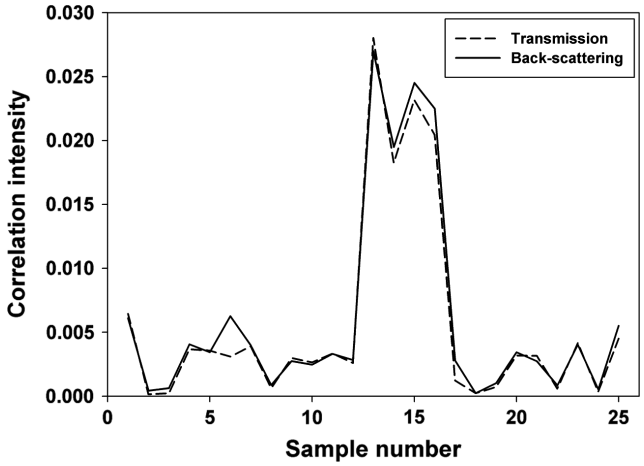

Figure 11. Auto-power spectra extracted using synchronous sample-to-sample 2D correlation analysis from backscattering (solid line) and transmission Raman measurement (dashed line) datasets.

in the transmission measurement are a consequence of successful sample representation of the whole pellet (Fig. 7). As stated, the narrower band implies a more rigid polymeric structure. As such, the transmission measurements should be more suitable for the correct sample representation of PE pellets.

Two-dimensional sample-to-sample correlation analysis

To further investigate the origin of the improved accuracy achieved using transmission measurements, both transmission and backscattering spectral datasets collected from the same 25 PE samples were analyzed using sample-to-sample 2D correlation analysis,[18–20] which is able to examine the similarity among samples in a given dataset. In the 2D map of sample-to-sample correlation, each point represents the correlation between a given pair of sample traces. The intensity of peaks located at diagonal positions mathematically corresponds to the autocorrelation function of spectral intensity variations. The diagonal peaks are therefore referred to as auto-peaks, and the slice trace of a synchronous sample-to-sample 2D spectrum along the diagonal is called the auto-power spectrum. Figure 11 shows the auto-power spectra extracted along the diagonal line on the synchronous sample-to-sample 2D correlation spectrum from backscattering (solid line) and transmission Raman measurement (dashed line) datasets. Auto-power spectrum corresponding to the synchronous correlation intensity at the main diagonal position in the synchronous sample-to-sample 2D correlation spectrum shows a much more vivid plot than the 2D contour plot of synchronous sample-to-sample 2D correlation analysis. The auto-power spectrum is conceptually identical to the classical spectrum-matching technique based on the calculation of spectral vector inner products. The correlation intensity becomes significant if the sample spectra are similar in shape. On the other hand, if the spectral shapes are substantially different, it approaches zero. In both cases, the correlation intensities of samples 13 to 16 were quite high compared to the other samples, indicating that the characteristics of these four samples are quite similar each other, while largely different from other 21 samples. As found in the auto-power spectra, these four samples were high-density PE samples, which are largely different from other copolymers in composition.

The correlation intensities calculated for each sample from both backscattering and transmission spectra were generally similar. However, the correlation intensities for large numbers of the samples from the transmission measurements are slightly lower than those from the backscattering measurements. This indicates that the samples were more spectroscopically discernable (or different) when the transmission measurement was used, even though the same samples were measured. The improved sample representation by transmission measurement helps in improving the spectral selectivity among the PE samples. Especially, for above-mentioned four high-density samples, the correlation intensities calculated from the transmission spectra are more substantially decreased compared to those obtained from the corresponding backscattering spectra. Since inhomogeneity of the pellet was more apparent for the high-density PEs as shown in Figs 8 and 9, the transmission measurement was more effective in improving the representation of a whole pellet and following the density variation more correctly.

Conclusions

Transmission Raman spectroscopy has been successful for the direct bulk analysis of packed PE pellets. Importantly, this study revealed that inhomogeneity inside a pellet can be clearly identified utilizing Raman line mapping and DSC analysis. The information presented in this paper, including pellet inhomogeneity and the necessity of correct sample representation, is truly significant in the field of polymer analysis. Since there are many different types of polymers with pellet forms, simple and nondestructive transmission Raman measurements ensuring correct sample representation will be beneficial for the analysis of diverse properties in combination with chemometric methods.

Diffuse reflectance NIR spectroscopy has been mostly utilized for the analysis of solid granular or pellet samples in polymers, agriculture, and feed industry. As with backscattering Raman measurements, diffuse reflectance NIR measurements could also produce spectral features more weighted by chemical information pertinent to the sample surface rather than the whole sample composition. In the near future, taking advantage of selective Raman spectral features as well as the capability of correct sample representation, transmission Raman measurements would gradually occupy the realm of diffuse reflectance NIR measurements in diverse fields of nondestructive analysis.

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